

Project Proposal:

Adapting Sparse Communication Graph Learning for Multi-Agent Coordination in Bandwidth-Constrained Settings

Ansh Dabral (ad258) Aneesh Thatiparti (vt20) Madhu Thatiparti (mt180)

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Project Type: Addressing a GraphML problem in a new setting.

1 Motivation

Cooperative multi-agent systems — from warehouse robot fleets to satellite constellations — must coordinate decisions across distributed agents that cannot observe the full world state. Graph Neural Networks (GNNs) have emerged as the natural architecture for this: agents are nodes and edges encode *who communicates with whom*. The dominant paradigm today is **Centralized Training, Decentralized Execution (CTDE)**, where communication graphs are designed or learned during training under the assumption of unlimited bandwidth.

However, this assumption breaks in real-world deployments. A team of search-and-rescue drones, a constellation of LEO satellites, or a fleet of autonomous vehicles must communicate over **noisy, bandwidth-limited channels**. Sending full hidden-state embeddings to all neighbors is infeasible. Yet most GraphML-based MARL methods are designed and evaluated under full-bandwidth conditions — a critical gap between research and deployment.

This project directly addresses this gap by taking a state-of-the-art sparse communication graph method — **ExpoComm** (ICLR 2025) [1] — which learns *who* to communicate with, and extending it to also learn *how much to compress* what is communicated, in a new bandwidth-constrained setting not studied in the original paper.

2 Problem Statement

Given: A team of N cooperative agents in a partially observable environment, where each agent can only communicate with neighbors over a channel with a hard bandwidth cap of B bits per timestep.

Goal: Learn simultaneously:

1. A sparse communication *topology* (which agents to communicate with), and
2. A compressed message *encoding* (what information to send within budget B).

The new setting: ExpoComm [1] was evaluated only on MAgent and IMP-MARL benchmarks. We apply it — combined with variational message compression from BVME [2] — to the **SMACv2** (StarCraft Multi-Agent Challenge v2) and **MPE** (Multi-Agent Particle Environments) benchmarks under varying levels of bandwidth constraint, settings not studied in either base paper.

3 Proposed Method

We propose **ExpoComm-B**: a modification of ExpoComm [1] that integrates the Bandwidth-constrained Variational Message Encoding (BVME) module from [2].

Component	Source	Our Contribution
Sparse topology (exponential graph)	ExpoComm [1]	Use directly
Variational compression (BVME module)	BVME [2]	Integrate into ExpoComm
Evaluation on SMACv2 + MPE under budget B	Neither paper	Our contribution

The key insight: ExpoComm decides *who* to talk to; BVME decides *what* to say efficiently. Neither paper combines both. We combine them and evaluate on new benchmarks.

Architecture overview:

Agent observation $\rightarrow$ GNN encoder (ExpoComm sparse topology) $\rightarrow$ BVME compression (bandwidth B) $\rightarrow$ Sparse message to selected neighbors $\rightarrow$ Aggregated messages $\rightarrow$ local policy $\rightarrow$ action

The BVME module treats each outgoing message as a sample from a learned Gaussian posterior, regularized via KL divergence. This provides principled, tunable control over compression strength through an interpretable bandwidth hyperparameter B .

4 Related Work

1. **ExpoComm** — Exponential Topology-enabled Scalable Communication in Multi-Agent Reinforcement Learning. Li et al., ICLR 2025.
<https://arxiv.org/abs/2502.19717> *Public code:* <https://github.com/LXXXXR/ExpoComm>
Base model. Learns sparse exponential communication graphs. Evaluated on MAgent and IMP-MARL.
2. **BVME** — Bandwidth-constrained Variational Message Encoding for Cooperative MARL. Duan et al., AAMAS 2026 (oral).
<https://arxiv.org/abs/2512.11179>
Source of the compression module. Evaluated on SMACv1/v2 and MPE. Reduces message dimensions by 67–83% with minimal performance loss.
3. **TGCNet** — Dynamic Directed Graph-Based Communication in Cooperative Multi-Agent Systems. Zhang et al., AAAI 2025.
<https://arxiv.org/abs/2408.07397>
Dynamic directed communication graph baseline for comparison.
4. **DG-MAPPO** — Multi-Agent Deep Reinforcement Learning Under Constrained Communications. Shaik et al., 2026.
<https://arxiv.org/abs/2601.17069>
Most recent fully distributed graph attention MARL baseline; tested on SMACv2, Google Football, MuJoCo.
5. **Factored Value Functions for Graph-Based MARL**. Rashwan et al., 2026.
<https://arxiv.org/abs/2601.11401>

- Graph diffusion-based credit assignment in large-scale MARL.*
6. **Mean-Field Control on Sparse Graphs.** Schmidt & Cui, 2026.
<https://arxiv.org/abs/2601.21477>
Theoretical justification for GNN-based actor-critics in sparse multi-agent settings.
 7. **Reinforcement Learning on Graphs: A Survey.** Nie et al., IEEE TETCI 2023.
<https://arxiv.org/abs/2204.06127>
Foundational taxonomy of graph RL methods and applications.
 8. **GNN Meets MARL: Fundamentals, Applications, and Future Directions.** Liu et al., 2024.
<https://arxiv.org/abs/2404.04898>
Comprehensive overview of GNN architectures used in cooperative MARL.

5 Datasets and Environments

Environment	Justification
SMACv2 (StarCraft II)	Standard MARL benchmark; used by BVME — enables direct comparison
MPE (Multi-Agent Particle)	Lightweight; used by both base papers; fast iteration for early experiments
MAgent (via ExpoComm code)	Already in ExpoComm codebase; enables large-scale tests (25–100 agents)

All three environments are open-source and widely used in MARL research. No custom datasets are required.

6 Experiments and Analysis Plan

We compare **four configurations** across all environments:

Method	Description
Full-comm baseline	All agents communicate with all neighbors (standard CTDE)
ExpoComm (original)	Sparse topology, no bandwidth limit
BVME only	Fixed full topology, variational compression only
ExpoComm-B (ours)	Sparse topology + variational compression (combined)

Metrics:

- Task reward (win rate / episodic return)
- Communication overhead (average bits transmitted per timestep)
- Scalability: performance vs. number of agents $N \in \{5, 10, 20\}$
- Zero-shot generalization: train on $N=5$ agents, test on $N=10$

Ablations:

- Vary bandwidth cap B : measure U-shaped sensitivity (very low vs. moderate vs. high)
- Sparse vs. dense topology under same bandwidth budget
- Effect of BVME KL regularization strength on coordination quality

7 Project Timeline

Week	Dates	Milestone
1	Feb 17–23	Set up ExpoComm codebase (conda env); reproduce original results on MAgent
2	Feb 24–Mar 2	Set up SMACv2 and MPE; run ExpoComm baseline on new environments
3	Mar 3–9	Understand BVME code; integrate BVME module into ExpoComm pipeline
4	Mar 10–16	Run ExpoComm-B experiments under varying bandwidth caps B
5	Mar 17–23	Full comparison (all 4 methods) on SMACv2 and MPE
6	Mar 24–30	Scalability experiments ($N = 5 \rightarrow 10 \rightarrow 20$); zero-shot transfer tests
7	Mar 31–Apr 6	Ablations, sensitivity analysis, figure generation
8	Apr 7–13	Draft final report
9	Apr 14–20	Polish report; clean and upload code to GitHub
10	Apr 21–28	Final submission and presentation preparation

8 Team Contributions

Member	Primary Responsibility
Aneesh Thatiparti (vt20)	ExpoComm codebase setup, BVME integration, SMACv2 experiments
Ansh Dabral (ad258)	MPE environment setup, baseline comparisons, ablation studies
Madhu Thatiparti (mt180)	Scalability experiments, quantitative analysis, report writing

9 Expected Contributions

1. **New benchmark evaluation:** First application of ExpoComm to SMACv2 and MPE under hard bandwidth constraints.
2. **Novel architectural combination:** First integration of sparse topology learning (ExpoComm) with variational message compression (BVME).
3. **Scalability analysis:** Empirical study of how communication topology quality and compression degrade (or hold) as team size grows.
4. **Public code:** Full implementation released on GitHub for reproducibility.

Code repository: `github.com/[team-repo]/ExpoComm-B` (to be created)

References

- [1] Xinran Li, Xiaolu Wang, Chenjia Bai, Jun Zhang. *Exponential Topology-enabled Scalable Communication in Multi-agent Reinforcement Learning*. ICLR 2025. <https://arxiv.org/abs/2502.19717>

- [2] Wei Duan, Jie Lu, En Yu, Junyu Xuan. *Bandwidth-constrained Variational Message Encoding for Cooperative Multi-agent Reinforcement Learning*. AAMAS 2026. <https://arxiv.org/abs/2512.11179>
- [3] Zhuohui Zhang, Bin He, Bin Cheng, Gang Li. *Bridging Training and Execution via Dynamic Directed Graph-Based Communication in Cooperative Multi-Agent Systems*. AAAI 2025. <https://arxiv.org/abs/2408.07397>
- [4] Shahil Shaik, Jonathon M. Smereka, Yue Wang. *Multi-Agent Deep Reinforcement Learning Under Constrained Communications*. arXiv 2026. <https://arxiv.org/abs/2601.17069>
- [5] Ahmed Rashwan, Keith Briggs, Chris Budd, Lisa Kreusser. *Factored Value Functions for Graph-Based Multi-Agent Reinforcement Learning*. arXiv 2026. <https://arxiv.org/abs/2601.11401>
- [6] Tobias Schmidt, Kai Cui. *Mean-Field Control on Sparse Graphs: From Local Limits to GNNs via Neighborhood Distributions*. arXiv 2026. <https://arxiv.org/abs/2601.21477>
- [7] Mingshuo Nie, Dongming Chen, Dongqi Wang. *Reinforcement Learning on Graphs: A Survey*. IEEE Transactions on Emerging Topics in Computational Intelligence, 2023. <https://arxiv.org/abs/2204.06127>
- [8] Ziheng Liu, Jiayi Zhang, Enyu Shi, Zhilong Liu, Dusit Niyato, Bo Ai, Xuemin Shen. *Graph Neural Network Meets Multi-Agent Reinforcement Learning: Fundamentals, Applications, and Future Directions*. arXiv 2024. <https://arxiv.org/abs/2404.04898>